

OBJECTIVES.md

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Summary

This document outlines a high-performance **engineering specification for an institutional-grade NIFTY options trading system** designed to capitalize on intraday market movements. The framework operates on a **synchronized ecosystem of volatility, order flow, and market-maker positioning**, specifically using the relationship between Put-Call Ratio (PCR) and Implied Volatility to avoid traps and identify high-probability breakouts. Its technical architecture is built for **low-latency execution**, utilizing an event-driven ingestion layer and a sophisticated signal engine that interprets **institutional gamma hedging and short-covering accumulation**. Ultimately, the system enforces a **strict risk governance model** and a composite execution matrix to ensure that trades are only initiated when technical indicators and time-of-day filters align for a distinct quantitative edge.

Keywords: Options Trading Strategy, Quantitative Edge Indicators, Technical Architecture, Put-Call Ratio Analysis, Risk Governance Layer

Content

NIFTY Options Trading System - Master Engineering Specification
This document consolidates the quantitative strategy, architectural blueprints, and operational procedures for the institutional-grade intraday NIFTY options buying system.

1. Strategy & Quantitative Edge
The system treats volatility, order flow, and market-maker positioning as a synchronized ecosystem. It seeks composite signatures where multiple indicators align.

1.1 Volatility-OI Nexus
Core Principle: Avoid "IV-Crush Traps" where PCR is high but implied volatility (IV) is collapsing.
Breakout Profile: Moderate PCR (~1.0) + Rising IV + Increasing Call OI.
Trap Profile: High PCR (>1.15) + Collapsing IV.

1.2 Market-Maker Gamma Hedging
Call Wall: Strike with the largest call gamma; dealers are short calls here and must buy underlying as price breaches this level.
Put Wall: Symmetric level on the downside.
Gamma Flip: The "Zero-

Gamma” level marking a regime change (above is dampening, below is amplifying volatility).Trigger: Price crossing an OI wall in a negative-gamma zone.1.3 Lead-Lag & VWAP DivergenceDivergence: Spot flat + Call-premium VWAP up + PCR↑ + Call OI↓ (short covering accumulation).Concurrent Pressure: Long call entries require Price_CE > VWAP_CE AND Price_PE < VWAP_PE.

2. Technical ArchitectureThe system is decoupled into three layers to ensure low latency and prevent race conditions.2.1 Ingestion & Storage LayerProtocols: Upstox V3 WebSocket (Protobuf). Zero-lag event-driven data ingestion.Data Frequency: Accommodates NSE India’s ~3-minute OI update interval while maintaining 1-second visual responsiveness.Normalizer: High-frequency ticks are consolidated into 1-second buckets for signal processing.Storage: Real-time ticks are persistently stored in DuckDB (data/market_data_v3.duckdb) using an asynchronous buffer and flusher to minimize disk I/O.Symbol Engine: Bidirectional hash map linking exchange strings (e.g., NIFTY26JUN2524000CE) to Upstox instrument_key tokens.2.2 Feature & Signal EnginePCR Regime: <0.7 (bearish), 0.7-1.1 (Neutral), >1.1 (bullish). -** THINK LIKE institutional trader **:institutional traders write (sell) most options, they drive market directions:When PCR is High (>1.1): Big players are writing aggressive Put options. They expect the market to stay above those strike prices. This creates strong support and a bullish structure.When PCR is Low (<0.7): Big players are writing aggressive Call options. They expect the market to stay below those strike prices. This creates heavy resistance and a bearish structure. PCR > 1.1 => Bullish (Institutions are selling heavy puts/creating support) PCR < 0.7 => Bearish (Institutions are selling heavy calls/creating resistance)Formulating a Nifty Trading Framework: To trade Nifty effectively, use the institutional ranges based on Open Interest (OI) PCR:Identifying the Bearish Zone (PCR < 0.7)Market Status: Oversold or heavily capped by Call sellers.Trading Action: Avoid long positions; look for short setups near resistance.Identifying the Neutral Zone (PCR 0.75 - 1.0)Market Status: Range-bound consolidation.Trading Action: Deploy range-bound strategies like Iron Condors.Identifying the Bullish Zone (PCR 1.1 - 1.4)Market Status: Strong upward momentum backed by heavy Put writing.Trading Action: Look to buy on dips; avoid shorting the index.Identifying Extreme Reversals (PCR > 1.5)Market Status: Hyper-bullish or overbought.Trading Action: Profit booking zone; watch for sudden profit-taking drops.IV Trend: Slope of ATM IV over 15-minute windows.Momentum Index: Weighted score of PCR_slope, VWAP_gap, and GEX_flow.Master Execution Matrix: Final GO/NO-GO decision logic applying time-of-day filters.2.3 Execution & Risk LayerRisk Governance: 2% capital risk per trade, 10% hard stop-loss, 15% take-profit threshold with 5% trailing buffer.Network Watchdog: Liquidates near-zero gamma positions if data delivery gaps exceed 5 seconds.

3. Composite Execution Matrix

4. Operational Runbook
4.1 DeploymentSet up .env with UPSTOX_ACCESS_TOKEN. Run `python3 src/main.py smoke` for sanity check. Launch dashboard: `python3 src/main.py dashboard -port 8000`.
4.2 Time-of-Day Blocks
Opening Hour (9:15-10:15): High conviction required; noise from overnight hedges.
Midday Lull (11:30-13:30): Lean passive; avoid low-volume false signals.
Power Hour (14:30-15:30): Focus on Max Pain shifts and gamma pin effects.